The Warping of Space: The Bulk and Our Brane

In 1912 Einstein realized that if time can be warped by massive bodies, then space must also be warped. But despite the most intense mental struggle of his life, the full details of space warps long eluded him. From 1912 to late 1915 he struggled. Finally in November 1915, in a great Eureka moment, he formulated his "field equation of general relativity," which encapsulated all his relativistic laws including space warps.

Again, human technology was too puny for high-precision tests.[9] This time the needed improvements took sixty years, culminating in several key experiments. The one I liked best was led by Robert Reasenberg and Irwin Shapiro of Harvard. In 1976–77 they transmitted radio signals to two spacecraft in orbit around Mars. The spacecraft, called *Viking 1* and *Viking 2*, amplified the signals and sent them back to Earth, where their round-trip travel time was measured. As the Earth and Mars moved around the Sun in their orbits, the radio signals traversed paths that were changing. At first, the paths were far from the Sun, then they passed near the Sun, and then far again, as shown in the bottom half of Figure 4.3.